

Tensor Formats in Banach Spaces

Minimal Subspaces, Geometry,
and Variational Principles

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Preface

This monograph presents a unified and self-contained treatment of the mathematical foundations of low-rank tensor formats in Banach spaces. It synthesises and extends the results of four research papers by the author and his collaborators Wolfgang Hackbusch and Anthony Nouy, published between 2012 and 2023.

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Notation and Conventions

See the file `notation.tex` for the complete list of macros and symbols used throughout.

Part I

**Algebraic and Topological
Foundations**

Chapter 1

Introduction and Motivation

To be written

This chapter is a placeholder. Content will be added in subsequent revisions.

Chapter 2

Banach Space Prerequisites

This chapter collects the functional-analytic and differential-geometric background required throughout the monograph. The reader familiar with Banach space theory and infinite-dimensional differential geometry may proceed directly to Chapter 3, referring back as needed. All vector spaces are over the field $\mathbb{K}$, which is either $\mathbb{R}$ or $\mathbb{C}$; unless stated otherwise, $\mathbb{K} = \mathbb{R}$.

2.1 Normed spaces and Banach spaces

We begin by fixing notation and recalling the central notions of normed and Banach space theory that will be used throughout.

Basic definitions

Let X be a vector space over $\mathbb{K}$. A *norm* on X is a map $\|\cdot\| : X \rightarrow [0, \infty)$ satisfying:

(N1) $\|x\| = 0$ if and only if $x = 0$,

(N2) $\|\lambda x\| = |\lambda| \|x\|$ for all $\lambda \in \mathbb{K}$, $x \in X$,

(N3) $\|x + y\| \leq \|x\| + \|y\|$ for all $x, y \in X$.

The pair $(X, \|\cdot\|)$ is called a *normed space*. A sequence $(x_n)_{n \in \mathbb{N}}$ in X is a *Cauchy sequence* if $\|x_n - x_m\| \rightarrow 0$ as $n, m \rightarrow \infty$. A normed space in which every Cauchy sequence converges is called a *Banach space*.

Two norms $\|\cdot\|$ and $\|\cdot\|'$ on X are *equivalent* if there exist constants $c, C > 0$ such that $c\|x\| \leq \|x\|' \leq C\|x\|$ for all $x \in X$. Equivalent norms define the same topology, and hence the same Cauchy sequences, so they produce the same Banach space structure.

Example 2.1.1. Let $I \subset \mathbb{R}$ be a measurable set and $1 \leq p < \infty$. The Lebesgue space $L^p(I)$, equipped with the norm $\|f\|_{L^p} = \left(\int_I |f(x)|^p dx\right)^{1/p}$, is a Banach space. For

$p = \infty$ one uses the essential supremum norm. The case $p = 2$ gives a Hilbert space with inner product $\langle f, g \rangle = \int_I f(x) \overline{g(x)} dx$.

Example 2.1.2. For a bounded open set $\Omega \subset \mathbb{R}^n$, $1 \leq p < \infty$, and $k \in \mathbb{N}_0$, the Sobolev space $W^{k,p}(\Omega)$ consists of all functions $f \in L^p(\Omega)$ whose distributional derivatives $D^\alpha f$ of order $|\alpha| \leq k$ belong to $L^p(\Omega)$. Equipped with the norm

$$\|f\|_{W^{k,p}} = \left(\sum_{|\alpha| \leq k} \|D^\alpha f\|_{L^p}^p \right)^{1/p},$$

it is a Banach space. For $p = 2$ one writes $H^k(\Omega) = W^{k,2}(\Omega)$, which is a Hilbert space. These spaces appear as natural component spaces in high-dimensional tensor problems; see Example 4.4.7 and Section ??.

Continuous linear maps

Let X and Y be normed spaces. A linear map $A : X \rightarrow Y$ is *continuous* if and only if it is *bounded*, meaning

$$\|A\|_{Y \leftarrow X} := \sup_{x \neq 0} \frac{\|Ax\|_Y}{\|x\|_X} = \sup_{\|x\|_X \leq 1} \|Ax\|_Y < \infty.$$

The quantity $\|A\|_{Y \leftarrow X}$ is the *operator norm* of A . The space of all continuous linear maps from X to Y , denoted $\mathcal{L}(X, Y)$, is itself a normed space under the operator norm; it is a Banach space whenever Y is complete. We write $\mathcal{L}(X) := \mathcal{L}(X, X)$. The set of bounded linear automorphisms (bounded invertible maps with bounded inverse) is

$$\mathrm{GL}(X) := \{A \in \mathcal{L}(X) : A \text{ invertible}\}.$$

This is an open subset of $\mathcal{L}(X)$, and hence a Banach manifold modelled on $\mathcal{L}(X)$; see Example 2.4.3.

Dual spaces and reflexivity

The *topological dual* of a normed space $(X, \|\cdot\|)$ is $X^* := \mathcal{L}(X, \mathbb{K})$, equipped with the dual norm

$$\|\varphi\|_{X^*} := \sup_{\|x\| \leq 1} |\varphi(x)|. \quad (2.1)$$

The dual X^* is always a Banach space, regardless of whether X is complete. The *bidual* $X^{**} := (X^*)^*$ contains X as a canonically embedded subspace via the isometric embedding $\iota : X \rightarrow X^{**}$, $\iota(x)(\varphi) = \varphi(x)$. When ι is surjective, X is called *reflexive*. All L^p spaces with $1 < p < \infty$ are reflexive; L^1 and L^∞ are not.

Reflexivity plays an important role in the existence of best approximations. Recall that a closed set C in a normed space X is *proximal* if every point of X has at least one nearest point in C . In a reflexive Banach space, every closed convex set is proximal (see, e.g., [Hac19, p. 61]). This fact will be used extensively in Chapter 4 to prove the existence of best low-rank tensor approximations.

Convexity and smoothness

Two geometric properties of the norm are of particular importance for the Dirac–Frenkel variational principle in Chapter 10.

Definition 2.1.3. A normed space $(X, \|\cdot\|)$ is *strictly convex* (or *rotund*) if $\|x+y\|/2 < 1$ for all $x, y \in X$ with $\|x\| = \|y\| = 1$ and $x \neq y$.

Definition 2.1.4. A normed space $(X, \|\cdot\|)$ is *uniformly convex* if for every $\varepsilon > 0$ there exists $\delta > 0$ such that $\|x+y\|/2 \leq 1 - \delta$ whenever $\|x\|, \|y\| \leq 1$ and $\|x-y\| \geq \varepsilon$.

Every uniformly convex Banach space is reflexive and strictly convex (Clarkson’s theorem). In particular, L^p spaces with $1 < p < \infty$ are uniformly convex. Strict convexity guarantees uniqueness of metric projections onto closed convex sets, a property needed for the well-posedness of the Dirac–Frenkel equation.

Definition 2.1.5. A Banach space X is *smooth* if the limit $\lim_{t \rightarrow 0} (\|x+ty\| - \|x\|)/t$ exists for all $x, y \in X$ with $\|x\| = 1$.

Smoothness is the dual property to strict convexity in the following sense: X is smooth if and only if X^* is strictly convex, and X is strictly convex if and only if X^* is smooth.

2.2 Complemented subspaces and projections

The central geometric notion underlying all manifold structures in this monograph is that of a complemented (or split) subspace. Unlike the Hilbert space setting, complementability is a non-trivial condition in general Banach spaces.

Projections

A bounded linear map $P \in \mathcal{L}(X)$ is a *projection* if $P^2 = P$. In that case $\text{Im } P$ is a closed subspace, $\text{Ker } P$ is closed, and $X = \text{Im } P \oplus \text{Ker } P$ (as a direct sum of closed subspaces). The complementary projection is $\text{id}_X - P$.

Definition 2.2.1. Let X be a Banach space. A closed subspace $U \subset X$ is a *split subspace* (or *complemented subspace*) of X if there exists a closed subspace $W \subset X$ such

that $X = U \oplus W$. The pair (U, W) is then called a *pair of complementary subspaces*, and W is a (*topological*) *complement* of U in X . We write $P_{U \oplus W}$ for the unique continuous projection from X onto U along W .

The following proposition characterises split subspaces in terms of projections and provides three equivalent conditions that will each be used at different points of the text.

Proposition 2.2.2 ([FHHM15, Proposition 4.2]). *Let X be a Banach space and $U \subset X$ a closed subspace. The following conditions are equivalent:*

- (a) *U is a split subspace of X .*
- (b) *There exists $P \in \mathcal{L}(X)$ with $P^2 = P$ and $\text{Im } P = U$.*
- (c) *There exists $Q \in \mathcal{L}(X)$ with $Q^2 = Q$ and $\text{Ker } Q = U$.*

In condition (b), the projection P is none other than $P_{U \oplus W}$ for any choice of closed complement W of U ; in (c), one has $Q = P_{W \oplus U}$.

Proposition 2.2.3 ([FHHM15, Theorem 4.5]). *Let X be a Banach space. Every finite-dimensional subspace of X is a split subspace.*

In a Hilbert space, every closed subspace is complemented (the orthogonal complement provides the split). This is no longer true in general Banach spaces: by a classical result of Lindenstrauss and Tzafriri, a Banach space in which every closed subspace is complemented is necessarily isomorphic to a Hilbert space. The failure of complementability in non-Hilbert spaces is precisely what makes the immersion theorems of Chapter 8 non-trivial.

The nilpotent algebra associated to a splitting

The following result, central to the construction of local charts in Chapters 8 and 9, identifies a particularly well-behaved sub-algebra of $\mathcal{L}(X)$ associated to any splitting $X = U \oplus W$.

Proposition 2.2.4. *Let X be a Banach space and $U, W \in \mathbb{G}(X)$ with $X = U \oplus W$. Define*

$$\mathcal{L}_{(U,W)}(X) := \{P_{W \oplus U} \circ S \circ P_{U \oplus W} : S \in \mathcal{L}(X)\} \cong \mathcal{L}(U, W).$$

Then:

- (i) *The natural map $\mathcal{L}(U, W) \rightarrow \mathcal{L}_{(U,W)}(X)$, $L \mapsto P_{W \oplus U} \circ L \circ P_{U \oplus W}$, is an isometric isomorphism.*

(ii) $\mathcal{L}_{(U,W)}(X)$ is a nilpotent sub-algebra of $\mathcal{L}(X)$: for every $L, L' \in \mathcal{L}_{(U,W)}(X)$ one has $L \circ L' = 0$.

(iii) As a consequence, for every $L \in \mathcal{L}_{(U,W)}(X)$:

$$\exp(L) := \sum_{n=0}^{\infty} \frac{L^n}{n!} = \text{id}_X + L, \quad \exp(-L) = \text{id}_X - L = (\text{id}_X + L)^{-1}. \quad (2.2)$$

Proof. For the isometry in (i), note that $\|P_{W \oplus U} \circ L \circ P_{U \oplus W}\|_{X \leftarrow X} = \|L\|_{W \leftarrow U}$ by the definition of the operator norm. For (ii): if $L = P_{W \oplus U} \circ S \circ P_{U \oplus W}$ and $L' = P_{W \oplus U} \circ S' \circ P_{U \oplus W}$, then

$$L \circ L' = P_{W \oplus U} \circ S \circ (P_{U \oplus W} \circ P_{W \oplus U}) \circ S' \circ P_{U \oplus W} = 0,$$

since $P_{U \oplus W} \circ P_{W \oplus U} = 0$. Statement (iii) follows immediately from (ii) and the series definition of the exponential. $\square$

The identity $\exp(L) = \text{id}_X + L$ means that the exponential map on the nilpotent algebra $\mathcal{L}_{(U,W)}(X)$ is a polynomial of degree one, and in particular a global analytic diffeomorphism. This is the key that makes the local charts of the Grassmann–Banach manifold and the Tucker manifold globally well-defined; see Section 2.3.

Common complements and chart compatibility

The following lemma, which underlies the compatibility of overlapping charts in the Grassmann–Banach manifold, characterises when two split subspaces share a common complement.

Lemma 2.2.5 ([DR63, Lemma 2.1]). *Let X be a Banach space and let $U, U', W \in \mathbb{G}(X)$ with $X = U \oplus W = U' \oplus W$. The following conditions are equivalent:*

- (a) U and U' have W as a common complement, i.e., $X = U \oplus W = U' \oplus W$.
- (b) The restriction $P_{U \oplus W}|_{U'} : U' \rightarrow U$ is a linear isomorphism (with bounded inverse).

Moreover, in this case $(P_{U \oplus W}|_{U'})^{-1} = P_{U' \oplus W}|_U$.

2.3 The Grassmann–Banach manifold

The Grassmannian of a Banach space is the natural parameter space for subspaces, and it serves as the base manifold for the Tucker and tree-based tensor manifolds.

Definition 2.3.1. Let X be a Banach space. The *Grassmann manifold* (or *Grassmannian*) of X is

$$\mathbb{G}(X) := \{U \subset X : U \text{ is a split subspace of } X\}.$$

For $r \in \mathbb{N}_0$, the *finite Grassmannian* is

$$\mathbb{G}_r(X) := \{U \in \mathbb{G}(X) : \dim U = r\}.$$

Both X and $\{0\}$ belong to $\mathbb{G}(X)$. By Proposition 2.2.3, every r -dimensional subspace of X belongs to $\mathbb{G}(X)$, so $\mathbb{G}_r(X)$ is well defined.

Theorem 2.3.2 (Grassmann–Banach manifold, [Dou66]). *Let X be a Banach space. The Grassmannian $\mathbb{G}(X)$ is an analytic Banach manifold, with atlas $\{\mathbb{G}(W, X), \Psi_{U \oplus W}\}_{U \in \mathbb{G}(X)}$, where for each splitting $X = U \oplus W$:*

(i) $\mathbb{G}(W, X) := \{V \in \mathbb{G}(X) : X = V \oplus W\}$ *is the chart domain at U .*

(ii) *The chart map $\Psi_{U \oplus W} : \mathbb{G}(W, X) \rightarrow \mathcal{L}(U, W)$ is defined by*

$$\Psi_{U \oplus W}(V) := P_{W \oplus U}|_V \circ (P_{U \oplus W}|_V)^{-1}, \quad (2.3)$$

with inverse $\Psi_{U \oplus W}^{-1}(L) = G(L) := \{(\text{id}_X + L)(u) : u \in U\}$.

The transition maps are analytic. Each chart domain $\mathbb{G}(W, X)$ is a Banach manifold modelled on $\mathcal{L}(U, W)$.

The chart map $\Psi_{U \oplus W}$ sends a subspace $V \in \mathbb{G}(W, X)$ to the linear map $L : U \rightarrow W$ whose graph $G(L)$ equals V . The identity $\Psi_{U \oplus W}^{-1}(L) = \exp(L)(U)$, combined with Proposition 2.2.4, shows that the inverse chart is a degree-one polynomial map (hence analytic). Analyticity of the transition maps follows by a direct computation using Lemma 2.2.5.

Remark 2.3.3. The full Grassmannian $\mathbb{G}(X)$ is a Banach manifold but is *not* modelled on a single Banach space: if U and U' are finite-dimensional with $\dim U \neq \dim U'$, then $\mathcal{L}(U, W)$ and $\mathcal{L}(U', W')$ are non-isomorphic. This is not a deficiency but a structural fact: $\mathbb{G}(X)$ decomposes into connected components $\mathbb{G}(X) = \bigsqcup_{r \in \mathbb{N}_0} \mathbb{G}_r(X)$, each of which is modelled on a single Banach space.

Proposition 2.3.4. *For each $r \in \mathbb{N}_0$, the finite Grassmannian $\mathbb{G}_r(X)$ is a connected component of $\mathbb{G}(X)$, and it is an analytic Banach manifold modelled on $\mathcal{L}(U, W)$ for any $U \in \mathbb{G}_r(X)$ and complement W (the isomorphism class of $\mathcal{L}(U, W)$ does not depend on the choice of U or W).*

Remark 2.3.5 (Grassmannian of a normed space). The construction extends to normed (non-complete) spaces. Let $(X, \|\cdot\|)$ be a normed space with completion $\bar{X}$. Define $\mathbb{G}_r(X)$

as the set of r -dimensional subspaces of X that are also split in $\bar{X}$. By Lemma 2.5.6, every chart domain $\mathbb{G}(W, \bar{X})$ is contained in $\mathbb{G}_r(X)$, so the atlas of $\mathbb{G}_r(\bar{X})$ restricts to an analytic atlas on $\mathbb{G}_r(X)$. This variant is needed in Section ??, where the component spaces V_α are normed but not necessarily complete.

The tangent space to $\mathbb{G}(X)$ at U is identified as follows.

Proposition 2.3.6. *For $U \in \mathbb{G}(X)$ with complement W , the tangent space $T_U \mathbb{G}(X) \cong \mathcal{L}(U, W)$.*

This identification is consistent with the finite-dimensional case: in $\mathbb{R}^n$, the tangent space to the Grassmannian $Gr(r, n)$ at a subspace U is the space of linear maps from U to its orthogonal complement, recovering the classical description.

2.4 Banach manifolds

We recall the definition of a Banach manifold in its general form, since the full Grassmannian $\mathbb{G}(X)$ and the full tensor space $\mathbf{V}_{D, \|\cdot\|_D}$ are manifolds not modelled on a fixed Banach space.

Definition 2.4.1. Let $\mathbb{M}$ be a set. An *atlas of class $\mathcal{C}^p$* (for $p \geq 0$ or $p = \infty$ or $p = \omega$, the analytic case) on $\mathbb{M}$ is a family of *charts* $\{(M_i, u_i)\}_{i \in A}$ satisfying:

(AT1) $\{M_i\}_{i \in A}$ covers $\mathbb{M}$.

(AT2) For each $i \in A$, the chart $u_i : M_i \rightarrow U_i$ is a bijection onto an open set U_i in some Banach space X_i , and for any $i, j \in A$, the set $u_i(M_i \cap M_j)$ is open in X_i .

(AT3) Whenever $M_i \cap M_j \neq \emptyset$, the *transition map* $u_j \circ u_i^{-1} : u_i(M_i \cap M_j) \rightarrow u_j(M_i \cap M_j)$ is a diffeomorphism of class $\mathcal{C}^p$.

Two atlases are *compatible* if their union is again an atlas. A $\mathcal{C}^p$ -*Banach manifold* is a set $\mathbb{M}$ together with a maximal (complete) atlas of class $\mathcal{C}^p$. If all X_i are isomorphic to a single Banach space X , we say $\mathbb{M}$ is *modelled on X* .

Example 2.4.2. Every Banach space X is a $\mathcal{C}^\infty$ -Banach manifold modelled on itself, with the single chart (X, id_X) .

Example 2.4.3. If X is a Banach space, then $\text{GL}(X)$ is an open subset of $\mathcal{L}(X)$, and hence an analytic Banach manifold modelled on $\mathcal{L}(X)$. The maps $A \mapsto A^{-1}$ and $(A, B) \mapsto AB$ are analytic (see [Upm85, 2.7, 2.8]).

Definition 2.4.4. Let $\mathbb{M}$ be a $\mathcal{C}^p$ -Banach manifold with $p \geq 1$ and $m \in \mathbb{M}$. A *tangent vector* at m is an equivalence class of triples (U_i, u_i, v) where (U_i, u_i) is a chart at m and

$v \in X_i$, under the equivalence $(U_i, u_i, v) \sim (U_j, u_j, w)$ if $D(u_j \circ u_i^{-1})(u_i(m))v = w$. The *tangent space* $T_m\mathbb{M}$ is the set of all tangent vectors at m , which inherits the structure of a Banach space from any chart.

Definition 2.4.5. A $\mathcal{C}^1$ -map $f : \mathbb{M} \rightarrow \mathbb{N}$ between Banach manifolds is an *immersion* at m if $Df(m) : T_m\mathbb{M} \rightarrow T_{f(m)}\mathbb{N}$ is injective with closed split image. It is an *immersion* if it is an immersion at every point. An *immersed submanifold* of $\mathbb{N}$ is the image of an injective immersion.

The closedness and split condition on the image of $Df(m)$ is automatic in finite dimensions and in Hilbert spaces (where every closed subspace is split), but is a genuine constraint in Banach spaces. This is the key difficulty resolved in Section ??.

2.5 Multilinear maps and Fréchet differentiability

Since the tensor product map is multilinear, the following general results will be used repeatedly.

Definition 2.5.1. A map $M : X_1 \times \cdots \times X_d \rightarrow Y$ between normed spaces is *multilinear* if it is linear in each argument separately. It is *bounded* (equivalently, *continuous*) if

$$\|M\| := \sup_{\substack{x_j \in X_j \\ \|x_j\|_j \leq 1}} \|M(x_1, \dots, x_d)\|_Y < \infty.$$

The following proposition shows that continuity implies a much stronger regularity.

Proposition 2.5.2 ([HN12, Proposition 79]). *Let $X_1, \dots, X_d$ and Y be normed spaces and let $M : X_1 \times \cdots \times X_d \rightarrow Y$ be a continuous multilinear map. Then M is $\mathcal{C}^\infty$ -Fréchet differentiable, and $D^k M = 0$ for all $k > d$.*

The vanishing of derivatives beyond order d reflects the polynomial nature of multilinear maps. In the complex case, a stronger conclusion holds.

Proposition 2.5.3 ([HN12, Theorem 160]). *Let X, Y be complex Banach spaces and $U \subset X$ open. A map $f : U \rightarrow Y$ is analytic if and only if it is $\mathcal{C}^1$ -Fréchet differentiable.*

Corollary 2.5.4. *Every continuous multilinear map between complex Banach spaces is analytic.*

In real Banach spaces, $\mathcal{C}^\infty$ and analytic are distinct notions; the tensor product map is $\mathcal{C}^\infty$ but need not be analytic over $\mathbb{R}$. For the real case, $\mathcal{C}^\infty$ -regularity is sufficient for the manifold constructions in Chapters 8 and 9.

Remark 2.5.5. Proposition 2.5.2 is the reason why the Tucker and tree-based tensor manifolds are $\mathcal{C}^\infty$ -Banach manifolds (and analytic over $\mathbb{C}$): their chart transition maps are compositions of continuous multilinear maps with bounded linear projections, all of which are $\mathcal{C}^\infty$ -Fréchet differentiable.

The Grassmannian revisited: Lemma on normed spaces

For later use in Section ??, we record the following lemma about Grassmannians of normed (non-complete) spaces.

Lemma 2.5.6. *Let $(X, \|\cdot\|)$ be a normed space with completion $\bar{X}$. Let $U \in \mathbb{G}_r(\bar{X})$ with $U \subset X$, and suppose $\bar{X} = U \oplus W$ for some $W \in \mathbb{G}(\bar{X})$. Then every $V \in \mathbb{G}(W, \bar{X})$ satisfies $V \subset X$.*

Proof. Observe that $X = U \oplus (W \cap X)$, where $W \cap X$ is dense in W . Suppose for contradiction that there exists $V \in \mathbb{G}(W, \bar{X})$ with $V \not\subset X$. Since V is finite-dimensional and $\bar{X} = V \oplus W$, the subspace $V \cap X$ is a proper subspace of V . On the other hand, $X = (V \cap X) \oplus (W \cap X)$ implies $\bar{X} = (V \cap X) \oplus W$, so $V \cap X \in \mathbb{G}(W, \bar{X})$. But $\dim(V \cap X) < \dim V = r$, contradicting the fact that every element of $\mathbb{G}(W, \bar{X})$ has dimension r . $\square$

Chapter 3

Algebraic Tensor Spaces

This chapter develops the purely algebraic foundations of tensor spaces. We introduce the algebraic tensor product via its universal property, establish the key notions of elementary tensor, matricisation, and multilinear rank, and define the Tucker format and the set $\mathfrak{M}_t(\mathbf{V}_D)$ of tensors of fixed Tucker rank. No topology is assumed in this chapter; that enters in Chapter 7.

Throughout, $D = \{1, \dots, d\}$ with $d \geq 2$ is the *dimension index set*, and V_α ($\alpha \in D$) are vector spaces over $\mathbb{K}$. Concerning the foundational construction of tensor products we follow Greub [Gre81].

3.1 The algebraic tensor product

Universal property and construction

Definition 3.1.1. The *algebraic tensor product* of the spaces V_α , $\alpha \in D$, is a pair $(\mathbf{V}_D, \otimes)$ consisting of a vector space $\mathbf{V}_D$ and a multilinear map

$$\otimes : \prod_{\alpha \in D} V_\alpha \longrightarrow \mathbf{V}_D, \quad (v_\alpha)_{\alpha \in D} \mapsto \bigotimes_{\alpha \in D} v_\alpha, \quad (3.1)$$

satisfying the following *universal property*: for every vector space Z and every multilinear map $M : \prod_{\alpha \in D} V_\alpha \rightarrow Z$, there exists a unique linear map $\widehat{M} : \mathbf{V}_D \rightarrow Z$ such that $M = \widehat{M} \circ \otimes$.

We write $\mathbf{V}_D = {}_a \otimes_{\alpha \in D} V_\alpha$ and refer to elements of the form $\bigotimes_{\alpha \in D} v_\alpha$ (with $v_\alpha \in V_\alpha$) as *elementary tensors*. Every element of $\mathbf{V}_D$ is a *finite* linear combination of elementary tensors.

The algebraic tensor product exists and is unique up to isomorphism; see [Gre81] for the standard construction. The prefix ‘*a*’ in ${}_a \otimes$ emphasises the algebraic (non-topological) nature. The underlying field is $\mathbb{K} = \mathbb{R}$ throughout, but all results hold equally for $\mathbb{K} = \mathbb{C}$.

Remark 3.1.2. The set of elementary tensors is *not* a basis of $\mathbf{V}_D$; it generates $\mathbf{V}_D$ but with many linear dependencies. For example, $\lambda v \otimes w = v \otimes \lambda w$ for all $\lambda \in \mathbb{K}$. If $\{e_i\}$ and $\{f_j\}$ are bases of V_1 and V_2 respectively, then $\{e_i \otimes f_j\}$ is a basis of $V_1 \otimes_a V_2$.

Bilinear identification and the α -unfolding

For each $\alpha \in D$, the algebraic tensor space can be rewritten as a two-factor product by grouping all indices except α :

$$\mathbf{V}_D \cong V_\alpha \otimes_a \mathbf{V}_\alpha^{[\alpha]}, \quad \mathbf{V}_\alpha^{[\alpha]} := {}_a \bigotimes_{\beta \in D \setminus \{\alpha\}} V_\beta. \quad (3.2)$$

Concretely, there is a canonical linear isomorphism $\Phi_\alpha : \mathbf{V}_D \xrightarrow{\sim} V_\alpha \otimes_a \mathbf{V}_\alpha^{[\alpha]}$ and throughout this monograph we identify $\mathbf{v} \in \mathbf{V}_D$ with $\Phi_\alpha(\mathbf{v}) \in V_\alpha \otimes_a \mathbf{V}_\alpha^{[\alpha]}$ without further comment. An elementary tensor splits as $\bigotimes_{\beta \in D} v_\beta \leftrightarrow v_\alpha \otimes \bigotimes_{\beta \neq \alpha} v_\beta$.

Tensor products of linear maps

Given linear maps $A_\alpha : V_\alpha \rightarrow W_\alpha$ for $\alpha \in D$, their tensor product is the unique linear map

$$\bigotimes_{\alpha \in D} A_\alpha : \mathbf{V}_D \longrightarrow \mathbf{W}_D := {}_a \bigotimes_{\alpha \in D} W_\alpha, \quad (3.3)$$

determined on elementary tensors by $\left(\bigotimes_{\alpha \in D} A_\alpha\right)\left(\bigotimes_{\alpha \in D} v_\alpha\right) = \bigotimes_{\alpha \in D} A_\alpha v_\alpha$. This is the unique linear extension guaranteed by the universal property. We denote the algebraic dual of V by $V' = L(V, \mathbb{K})$ and the space of all linear maps from V to W by $L(V, W)$.

Partitions and iterated tensor products

Definition 3.1.3. Let $\mathcal{P}$ be a partition of D (i.e., a collection of non-empty, pairwise disjoint subsets of D whose union is D). For each $\alpha \in \mathcal{P}$, set $\mathbf{V}_\alpha := {}_a \bigotimes_{j \in \alpha} V_j$ (with the convention $\mathbf{V}_{\{j\}} = V_j$). Then the *iterated tensor product* associated to $\mathcal{P}$ is

$${}_a \bigotimes_{\alpha \in \mathcal{P}} \mathbf{V}_\alpha \cong \mathbf{V}_D, \quad (3.4)$$

where the isomorphism is canonical and consistent with the identifications of elementary tensors.

This associativity-of-tensor-products fact is fundamental: it allows us to regroup the factors of $\mathbf{V}_D$ according to any hierarchical structure, which is precisely what the dimension partition trees of Chapter 5 will exploit.

Example 3.1.4. For $D = \{1, 2, 3, 4\}$ and the partition $\mathcal{P} = \{\{1, 2\}, \{3, 4\}\}$:

$$(V_1 \otimes_a V_2) \otimes_a (V_3 \otimes_a V_4) \cong V_1 \otimes_a V_2 \otimes_a V_3 \otimes_a V_4.$$

The partition $\mathcal{P}' = \{\{1\}, \{2\}, \{3\}, \{4\}\}$ (trivial) recovers the original four-fold product. The non-trivial partition $\mathcal{P}$ is the structure underlying the Tucker format (depth-2 tree), while more refined partitions correspond to hierarchical formats; see Chapter 5.

3.2 Matricisation and multilinear rank

Matricisation in finite dimensions

When the component spaces are finite-dimensional, the tensor $\mathbf{v} \in \mathbf{V}_D$ can be represented as a multi-dimensional array, and the identification (3.2) induces a matrix.

Definition 3.2.1. Let $D = \{1, \dots, d\}$, $\dim V_\alpha = n_\alpha < \infty$, and fix bases of each V_α . Identify $\mathbf{v} \in \mathbf{V}_D$ with its coordinate array $C \in \mathbb{R}^{n_1 \times \dots \times n_d}$. For each $\alpha \in D$, the α -matricisation (or α -unfolding) of $\mathbf{v}$ is the matrix

$$\mathcal{M}_\alpha(\mathbf{v}) \in \mathbb{R}^{n_\alpha \times \prod_{\beta \neq \alpha} n_\beta}$$

obtained by reordering the coordinate array C so that the α -index labels the rows and all other indices, in lexicographic order, label the columns. The rank of this matrix, $\text{rank} \mathcal{M}_\alpha(\mathbf{v}) \in \{0, 1, \dots, n_\alpha\}$, is the α -rank of $\mathbf{v}$.

The notion of α -rank was introduced by Hitchcock [Hit27, p.170] in 1927 under the name “rank on the j th index”. In the infinite-dimensional case, the correct replacement of $\text{rank} \mathcal{M}_\alpha(\mathbf{v})$ is the dimension of the minimal subspace $U_\alpha^{\min}(\mathbf{v})$, which we define in Section 3.3 and develop fully in Chapter 4.

Full-rank core tensors

The following definition isolates the tensors whose coordinate arrays have maximal rank in every direction, a condition that will characterise elements of the Tucker manifold $\mathfrak{M}_\mathbf{r}(\mathbf{V}_D)$.

Definition 3.2.2. For a rank tuple $\mathbf{r} = (r_\alpha)_{\alpha \in D} \in \mathbb{N}^d$, define

$$\mathbb{R}_*^{\times_{\alpha \in D} r_\alpha} := \left\{ C \in \mathbb{R}^{\times_{\alpha \in D} r_\alpha} : \text{rank} \mathcal{M}_\alpha(C) = r_\alpha \text{ for all } \alpha \in D \right\}. \quad (3.5)$$

Elements of $\mathbb{R}_*^{\times_{\alpha \in D} r_\alpha}$ are called *full-rank core tensors* of shape $\mathbf{r}$.

Remark 3.2.3. Since $\text{rank } \mathcal{M}_\alpha(C) = r_\alpha$ if and only if $\mathcal{M}_\alpha(C)\mathcal{M}_\alpha(C)^T \in \text{GL}(\mathbb{R}^{r_\alpha})$, and the determinant is a continuous function, $\mathbb{R}_*^{\times_{\alpha \in D} r_\alpha}$ is an open subset of $\mathbb{R}^{\times_{\alpha \in D} r_\alpha}$, hence a smooth finite-dimensional manifold in its own right. In the scalar case $r_\alpha = 1$ for all α , we recover $\mathbb{R}_* = \mathbb{R} \setminus \{0\} = \text{GL}(\mathbb{R})$.

3.3 The Tucker format

Minimal subspaces: first appearance

We introduce minimal subspaces here in their algebraic form; the full theory (existence, uniqueness, lattice properties, best approximation) is developed in Chapter 4.

Definition 3.3.1. Let $\mathbf{v} \in \mathbf{V}_D = {}_a \bigotimes_{\alpha \in D} V_\alpha$ and $\alpha \in D$. Under the identification $\mathbf{V}_D \cong V_\alpha \otimes_a \mathbf{V}_\alpha^{[\alpha]}$, the *minimal subspace* of $\mathbf{v}$ in direction α is the unique subspace $U_\alpha^{\min}(\mathbf{v}) \subset V_\alpha$ of smallest dimension such that $\mathbf{v} \in U_\alpha^{\min}(\mathbf{v}) \otimes_a \mathbf{V}_\alpha^{[\alpha]}$. The α -rank of $\mathbf{v}$ is $r_\alpha(\mathbf{v}) := \dim U_\alpha^{\min}(\mathbf{v})$.

The key properties — existence, uniqueness, and the symmetry $\dim U_\alpha^{\min}(\mathbf{v}) = \dim U_{D \setminus \{\alpha\}}^{\min}(\mathbf{v})$ — are established in Chapter 4. Here we proceed taking these properties for granted.

The Tucker format and the set $\mathfrak{M}_\mathbf{r}(\mathbf{V}_D)$

Definition 3.3.2. Let $\mathbf{r} = (r_\alpha)_{\alpha \in D} \in \mathbb{N}_0^d$ and let $U_\alpha \in \mathbb{G}_{r_\alpha}(V_\alpha)$ for each $\alpha \in D$. A tensor $\mathbf{v} \in \mathbf{V}_D$ is said to be in *Tucker format* with subspace tuple $(U_\alpha)_{\alpha \in D}$ if

$$\mathbf{v} \in {}_a \bigotimes_{\alpha \in D} U_\alpha.$$

The *Tucker rank* of $\mathbf{v}$ is the tuple $\mathbf{r}(\mathbf{v}) = (r_\alpha(\mathbf{v}))_{\alpha \in D}$, and the set of tensors with fixed Tucker rank $\mathbf{r}$ is

$$\mathfrak{M}_\mathbf{r}(\mathbf{V}_D) := \left\{ \mathbf{v} \in \mathbf{V}_D : r_\alpha(\mathbf{v}) = r_\alpha \text{ for all } \alpha \in D \right\}. \quad (3.6)$$

The Tucker format thus represents $\mathbf{v}$ as an element of a subspace $U_1 \otimes \cdots \otimes U_d$ spanned by r_α -dimensional subspaces $U_\alpha \subset V_\alpha$. Unlike the subspaces, the representation is not unique; the set $\mathfrak{M}_\mathbf{r}(\mathbf{V}_D)$ collects all tensors for which the *minimal* such subspaces have dimension exactly r_α .

Admissible Tucker ranks

Definition 3.3.3. A tuple $\mathbf{r} = (r_\alpha)_{\alpha \in D} \in \mathbb{N}_0^d$ is an *admissible Tucker rank* for $\mathbf{V}_D$ if there exists $\mathbf{v} \in \mathbf{V}_D$ with $r_\alpha(\mathbf{v}) = r_\alpha$ for all $\alpha \in D$. The set of all admissible Tucker

ranks is

$$\mathcal{AD}(\mathbf{V}_D) := \left\{ (r_\alpha(\mathbf{v}))_{\alpha \in D} \in \mathbb{N}_0^d : \mathbf{v} \in \mathbf{V}_D \right\}. \quad (3.7)$$

The zero tuple $\mathbf{0} = (0, \dots, 0)$ is always admissible (corresponding to $\mathbf{v} = 0$). The tuple $\mathbf{1} = (1, \dots, 1)$ is admissible if and only if $\dim V_\alpha \geq 1$ for all α . Since the r_α are bounded above by $\dim V_\alpha$ (in finite dimensions), $\mathcal{AD}(\mathbf{V}_D)$ is a finite set when all V_α are finite-dimensional.

The following decomposition is one of the foundational structural facts of the whole theory.

Proposition 3.3.4. *The algebraic tensor space decomposes as a disjoint union:*

$$\mathbf{V}_D = \bigsqcup_{\mathfrak{r} \in \mathcal{AD}(\mathbf{V}_D)} \mathfrak{M}_{\mathfrak{r}}(\mathbf{V}_D). \quad (3.8)$$

Proof. Since every $\mathbf{v} \in \mathbf{V}_D$ has a well-defined Tucker rank tuple (Chapter 4), the sets $\mathfrak{M}_{\mathfrak{r}}(\mathbf{V}_D)$ cover $\mathbf{V}_D$; they are disjoint by definition of the rank. $\square$

The decomposition (3.8) is the algebraic precursor of the manifold decomposition established in Chapter 8: each component $\mathfrak{M}_{\mathfrak{r}}(\mathbf{V}_D)$ will be shown to carry the structure of an infinite-dimensional $\mathcal{C}^\infty$ -Banach manifold.

Explicit representation of Tucker tensors

The following lemma, which combines the definition of $\mathfrak{M}_{\mathfrak{r}}(\mathbf{V}_D)$ with the properties of minimal subspaces, gives three equivalent characterisations of Tucker tensors. It is the algebraic heart of the Tucker manifold construction.

Lemma 3.3.5. *Let $\mathbf{V}_D = {}_a \otimes_{\alpha \in D} V_\alpha$ and $\mathfrak{r} \in \mathcal{AD}(\mathbf{V}_D)$ with $\mathfrak{r} \neq \mathbf{0}$. For $\mathbf{v} \in \mathbf{V}_D$, the following conditions are equivalent.*

- (a) $\mathbf{v} \in \mathfrak{M}_{\mathfrak{r}}(\mathbf{V}_D)$.
- (b) (Tucker representation) *For each $\alpha \in D$ there exists a basis $\{u_{i_\alpha}^{(\alpha)} : 1 \leq i_\alpha \leq r_\alpha\}$ of $U_\alpha^{\min}(\mathbf{v})$ and a unique full-rank core tensor $C^{(D)} \in \mathbb{R}_*^{\times_{\alpha \in D} r_\alpha}$, once the bases are fixed, such that*

$$\mathbf{v} = \sum_{\substack{1 \leq i_\alpha \leq r_\alpha \\ \alpha \in D}} C_{(i_\alpha)_{\alpha \in D}}^{(D)} \bigotimes_{\alpha \in D} u_{i_\alpha}^{(\alpha)}. \quad (3.9)$$

- (c) (Bilinear form) *For each $\alpha \in D$ there exist linearly independent vectors $\{u_{i_\alpha}^{(\alpha)}\}_{i_\alpha=1}^{r_\alpha}$ in V_α and linearly independent vectors $\{\mathbf{U}_{i_\alpha}^{(\alpha)}\}_{i_\alpha=1}^{r_\alpha}$ in $\mathbf{V}_\alpha^{[\alpha]}$ such that*

$$\mathbf{v} = \sum_{i_\alpha=1}^{r_\alpha} u_{i_\alpha}^{(\alpha)} \otimes \mathbf{U}_{i_\alpha}^{(\alpha)}. \quad (3.10)$$

Moreover, when (3.9) holds, the vectors $\mathbf{U}_{i_\alpha}^{(\alpha)}$ are given by

$$\mathbf{U}_{i_\alpha}^{(\alpha)} = \sum_{\substack{1 \leq i_\beta \leq r_\beta \\ \beta \in D \setminus \{\alpha\}}} C_{i_\alpha, (i_\beta)_{\beta \neq \alpha}}^{(D)} \bigotimes_{\beta \neq \alpha} u_{i_\beta}^{(\beta)}. \quad (3.11)$$

Proof. (a) $\Leftrightarrow$ (c). If (a) holds, then by Proposition ?? (Chapter 4), $\mathbf{v} \in U_\alpha^{\min}(\mathbf{v}) \otimes_a U_{D \setminus \{\alpha\}}^{\min}(\mathbf{v})$ with $\dim U_\alpha^{\min}(\mathbf{v}) = \dim U_{D \setminus \{\alpha\}}^{\min}(\mathbf{v}) = r_\alpha$, which gives (c). Conversely, (c) implies $\dim U_\alpha^{\min}(\mathbf{v}) = r_\alpha$ for all α , hence (a).

(b) $\Rightarrow$ (c). Immediate.

(c) $\Rightarrow$ (b). From (c), one has $U_\alpha^{\min}(\mathbf{v}) = \text{span}\{u_{i_\alpha}^{(\alpha)}\}$ for each α , and since $\mathbf{v} \in {}_a \bigotimes_{\alpha \in D} U_\alpha^{\min}(\mathbf{v})$, there exists $C^{(D)} \in \mathbb{R}^{\times_{\alpha \in D} r_\alpha}$ satisfying (3.9). From (3.11) and the linear independence of the $\mathbf{U}_{i_\alpha}^{(\alpha)}$, one deduces $\text{rank} \mathcal{M}_\alpha(C^{(D)}) = r_\alpha$ for all α , so $C^{(D)} \in \mathbb{R}_*^{\times_{\alpha \in D} r_\alpha}$. $\square$

Remark 3.3.6. From Lemma 3.3.5 and Remark 3.2.3, once bases of the minimal subspaces $U_\alpha^{\min}(\mathbf{v})$ are fixed ($\alpha \in D$), there is a natural identification

$$\mathfrak{M}_\tau \left({}_a \bigotimes_{\alpha \in D} U_\alpha^{\min}(\mathbf{v}) \right) \xrightarrow{\sim} \mathbb{R}_*^{\times_{\alpha \in D} r_\alpha}, \quad \mathbf{u} \mapsto E^{(D)}, \quad (3.12)$$

where $E^{(D)}$ is the coordinate array of $\mathbf{u}$ in the chosen bases. This identification is a diffeomorphism (both directions are polynomial), and it is the local model for the Tucker manifold chart; see Section ??.

The Tucker format as a flat case of tree-based formats

The Tucker format corresponds to the *shallowest* possible hierarchical structure: the dimension partition tree T_D^{Tucker} of depth 1, whose root is D and whose children are the singletons $\{1\}, \dots, \{d\}$:

$$T_D^{\text{Tucker}} = \{D, \{1\}, \{2\}, \dots, \{d\}\}, \quad S(D) = \{\{1\}, \dots, \{d\}\} = \mathcal{L}(T_D)[T_D^{\text{Tucker}}].$$

In Chapter 6 we will define tree-based formats for arbitrary dimension partition trees T_D and show that $\mathfrak{M}_\tau(\mathbf{V}_D) = \mathcal{FT}_\tau(\mathbf{V}_D, T_D^{\text{Tucker}})$.

3.4 The tensor product map and its differential

When the component spaces are normed, the tensor product map inherits both topological and differential properties that underpin the manifold constructions in Part III.

Definition 3.4.1. Let $(V_\alpha, \|\cdot\|_\alpha)$ be normed spaces and let $\|\cdot\|_D$ be a norm on $\mathbf{V}_D$. The *tensor product map*

$$\bigotimes : \prod_{\alpha \in D} (V_\alpha, \|\cdot\|_\alpha) \longrightarrow (\mathbf{V}_D, \|\cdot\|_D), \quad (v_\alpha)_{\alpha \in D} \mapsto \bigotimes_{\alpha \in D} v_\alpha, \quad (3.13)$$

(where the product is equipped with the max norm $\|(v_\alpha)\|_\times = \max_\alpha \|v_\alpha\|_\alpha$) is said to be *continuous* if $\|\bigotimes_\alpha v_\alpha\|_D \leq C \prod_\alpha \|v_\alpha\|_\alpha$ for some constant $C > 0$.

Continuity of the tensor product map is equivalent to $\|\bigotimes\| < \infty$; it holds whenever $\|\cdot\|_D$ is a crossnorm (Definition 4.4.1) on $\mathbf{V}_D$.

Proposition 3.4.2. Let $(V_\alpha, \|\cdot\|_\alpha)$ be normed spaces and let $\|\cdot\|_D$ be a norm on $\mathbf{V}_D$ such that the tensor product map (3.13) is continuous. Then:

(i) The map (3.13) is $\mathcal{C}^\infty$ -Fréchet differentiable, with differential

$$D\bigotimes(v_1, \dots, v_d)(w_1, \dots, w_d) = \sum_{\alpha \in D} v_1 \otimes \dots \otimes v_{\alpha-1} \otimes w_\alpha \otimes v_{\alpha+1} \otimes \dots \otimes v_d. \quad (3.14)$$

(ii) The tensor product of bounded linear maps,

$$\bigotimes : \prod_{\alpha \in D} \mathcal{L}(U_\alpha, V_\alpha) \longrightarrow \mathcal{L}\left(\bigotimes_{\alpha \in D} U_\alpha, \mathbf{V}_D\right), \quad (A_\alpha) \mapsto \bigotimes_{\alpha \in D} A_\alpha,$$

is also continuous and $\mathcal{C}^\infty$ -Fréchet differentiable for any finite-dimensional subspaces $U_\alpha \subset V_\alpha$.

Proof. Part (i) follows directly from Proposition 2.5.2, since $\bigotimes$ is a continuous multilinear map. The formula for the differential follows by the Leibniz rule for multilinear maps. For (ii), one checks boundedness of the map using the continuity of (3.13):

$$\left\| \bigotimes_{\alpha} A_\alpha \right\|_{\mathbf{V}_D \leftarrow \bigotimes U_\alpha} \leq C \prod_{\alpha} \|A_\alpha\|_{V_\alpha \leftarrow U_\alpha},$$

where C is the constant in Definition 3.4.1, and the finite-dimensionality of U_α ensures $L(U_\alpha, V_\alpha) = \mathcal{L}(U_\alpha, V_\alpha)$. Fréchet differentiability then follows again from Proposition 2.5.2. $\square$

Remark 3.4.3. Proposition 3.4.2 is the reason why the Tucker manifold $\mathfrak{M}_t(\mathbf{V}_D)$ is $\mathcal{C}^\infty$ (not merely continuous): all coordinate change maps between charts are built from compositions of projections and the tensor product map, all of which are $\mathcal{C}^\infty$ -Fréchet differentiable. See Section ?? for the precise argument.

Banach tensor spaces

When a norm is fixed on $\mathbf{V}_D$, its completion is the natural ambient space.

Definition 3.4.4. Let $(V_\alpha, \|\cdot\|_\alpha)$ be normed spaces and $\|\cdot\|_D$ a norm on $\mathbf{V}_D = {}_a \bigotimes_{\alpha \in D} V_\alpha$. The *Banach tensor space* associated to this data is the completion

$$\mathbf{V}_{D, \|\cdot\|_D} := \overline{\|\cdot\|_D} \bigotimes_{\alpha \in D} V_\alpha := \overline{\mathbf{V}_D}^{\|\cdot\|_D}.$$

If $\mathbf{V}_{D, \|\cdot\|_D}$ is a Hilbert space, it is called a *Hilbert tensor space*.

Example 3.4.5. For $I_\alpha \subset \mathbb{R}$ ($\alpha \in D$) and $1 \leq p < \infty$, the Lebesgue spaces $L^p(I_\alpha)$ satisfy

$$L^p\left(\prod_{\alpha \in D} I_\alpha\right) = \|\cdot\|_{0,p} \bigotimes_{\alpha \in D} L^p(I_\alpha),$$

where $\|\cdot\|_{0,p}$ denotes the standard L^p norm on the product domain. For $p = 2$ this is a Hilbert tensor space.

Example 3.4.6. For Sobolev spaces $H^{N,p}(I_\alpha)$ with $1 < p < \infty$ and $\mathbf{I} = \prod_{\alpha} I_\alpha$:

$$H^{N,p}(\mathbf{I}) = \|\cdot\|_{N,p} \bigotimes_{\alpha \in D} H^{N,p}(I_\alpha).$$

This is reflexive and separable. For $p = 2$ it is a Hilbert tensor space. These spaces arise naturally in the numerical treatment of high-dimensional PDEs; see Section 4.4 and Section ??.

Remark 3.4.7. The algebraic tensor space $\mathbf{V}_D$ is dense in $\mathbf{V}_{D, \|\cdot\|_D}$ by construction. The Tucker manifold $\mathfrak{M}_t(\mathbf{V}_D)$ lives inside the algebraic space, but its closure (as a subset of $\mathbf{V}_{D, \|\cdot\|_D}$) and its topology as a manifold are determined by the norm $\|\cdot\|_D$ and, in particular, by whether $\|\cdot\|_D$ is a crossnorm. This interplay between algebraic and topological structure is a recurring theme throughout the monograph.

Chapter 4

Minimal Subspaces in Tensor Representations

In Chapter 3 we introduced the Tucker format and defined the minimal subspace $U_\alpha^{\min}(\mathbf{v})$ informally, taking its existence and properties for granted. This chapter provides the complete theory: existence and uniqueness (Section 4.1), the lattice structure and inclusion under refinement of partitions (Section 4.2), admissible ranks and the disjoint decomposition of $\mathbf{V}_D$ (Section 4.3), and the existence of best approximations in Banach tensor spaces (Section 4.4). The results in this chapter are taken from [FH12], expanded with complete proofs and additional examples.

4.1 Existence and uniqueness of minimal subspaces

The case $d = 2$: Schmidt decomposition

The case of two factors $\mathbf{V}_D = V_1 \otimes_a V_2$ is classical and serves as the inductive base. The key observation is a lattice property of tensor subspaces.

Lemma 4.1.1 (Intersection lemma). *Let X_i, Y_i be subspaces of V_i for $i = 1, 2$. Then*

$$(X_1 \otimes_a X_2) \cap (Y_1 \otimes_a Y_2) = (X_1 \cap Y_1) \otimes_a (X_2 \cap Y_2). \quad (4.1)$$

Proof. The inclusion $(X_1 \cap Y_1) \otimes_a (X_2 \cap Y_2) \subseteq (X_1 \otimes_a X_2) \cap (Y_1 \otimes_a Y_2)$ is immediate. For the reverse inclusion, let $\mathbf{v} \in (X_1 \otimes_a X_2) \cap (Y_1 \otimes_a Y_2)$ with representations

$$\mathbf{v} = \sum_{\nu=1}^{n_x} x_\nu^{(1)} \otimes x_\nu^{(2)} = \sum_{\nu=1}^{n_y} y_\nu^{(1)} \otimes y_\nu^{(2)}, \quad x_\nu^{(i)} \in X_i, y_\nu^{(i)} \in Y_i.$$

By adding terms with zero coefficient, assume $n_x = n_y = n$ and that $\{x_\nu^{(1)}\}$ and $\{y_\nu^{(1)}\}$ are each linearly independent (passing to a minimal representation). The two representations yield the same matrix $M_1(\mathbf{v})$ under any identification with a bilinear

form, so $\text{span}\{x_\nu^{(1)}\} = \text{span}\{y_\nu^{(1)}\} \subset X_1 \cap Y_1$. A symmetric argument for the second factor gives $\text{span}\{x_\nu^{(2)}\} \subset X_2 \cap Y_2$, so $\mathbf{v} \in (X_1 \cap Y_1) \otimes_a (X_2 \cap Y_2)$. $\square$

Proposition 4.1.2 (Minimal subspace: case $d = 2$). *Let $\mathbf{v} \in V_1 \otimes_a V_2$. There exist unique subspaces $U_i \subset V_i$ of finite dimension such that:*

(i) $\mathbf{v} \in U_1 \otimes_a U_2$.

(ii) If $\mathbf{v} \in X_1 \otimes_a X_2$ for any subspaces X_i , then $U_i \subset X_i$.

Moreover, $\dim U_1 = \dim U_2$.

Proof. Existence. Fix any representation $\mathbf{v} = \sum_\nu v_\nu^{(1)} \otimes v_\nu^{(2)}$ and set $U_i = \text{span}\{v_\nu^{(i)}\}$. By passing to a representation with linearly independent $v_\nu^{(1)}$, one can also take $v_\nu^{(2)}$ linearly independent (otherwise row reduce). Then $\dim U_1 = \dim U_2$ (both equal the rank of the bilinear form).

Minimality. If $\mathbf{v} \in X_1 \otimes_a X_2$, then by the intersection lemma, $\mathbf{v} \in (U_1 \cap X_1) \otimes_a (U_2 \cap X_2)$, which forces $U_i \subset X_i$ by minimality of U_i .

Uniqueness. If U_i and U'_i both satisfy (i)–(ii), minimality of U_i applied to U'_i gives $U_i \subset U'_i$, and vice versa. $\square$

The general case: existence and uniqueness

Theorem 4.1.3 (Existence and uniqueness of minimal subspaces). *Let $D = \{1, \dots, d\}$, $\mathbf{V}_D = {}_a \bigotimes_{\alpha \in D} V_\alpha$, and $\mathbf{v} \in \mathbf{V}_D$. For each $\alpha \in D$, under the identification $\mathbf{V}_D \cong V_\alpha \otimes_a \mathbf{V}_\alpha^{[\alpha]}$, there exists a unique subspace $U_\alpha^{\min}(\mathbf{v}) \subset V_\alpha$ of finite dimension $r_\alpha < \infty$ such that:*

(a) $\mathbf{v} \in U_\alpha^{\min}(\mathbf{v}) \otimes_a \mathbf{V}_\alpha^{[\alpha]}$.

(b) If $\mathbf{v} \in U_\alpha \otimes_a \mathbf{V}_\alpha^{[\alpha]}$ for any subspace $U_\alpha \subset V_\alpha$, then $U_\alpha^{\min}(\mathbf{v}) \subset U_\alpha$.

Furthermore, there exists a unique subspace $U_{D \setminus \{\alpha\}}^{\min}(\mathbf{v}) \subset \mathbf{V}_\alpha^{[\alpha]}$ of dimension r_α such that $\mathbf{v} \in U_\alpha^{\min}(\mathbf{v}) \otimes_a U_{D \setminus \{\alpha\}}^{\min}(\mathbf{v})$.

Proof. Apply Proposition 4.1.2 to the two-factor product $V_\alpha \otimes_a \mathbf{V}_\alpha^{[\alpha]}$. The subspace $U_\alpha^{\min}(\mathbf{v})$ is the minimal subspace in the V_α -factor. Uniqueness follows from the same argument as in $d = 2$. $\square$

Definition 4.1.4. The subspace $U_\alpha^{\min}(\mathbf{v})$ of Theorem 4.1.3 is called the *minimal subspace* of $\mathbf{v}$ in direction α . The integer $r_\alpha(\mathbf{v}) := \dim U_\alpha^{\min}(\mathbf{v})$ is the α -rank of $\mathbf{v}$.

Characterisation via contractions

The following characterisation, which is central to both the proof of Chapter 8 and the tree-based theory of Chapter 6, expresses $U_\alpha^{\min}(\mathbf{v})$ as the range of a family of contractions.

Theorem 4.1.5 (Contraction characterisation). *Let $\mathbf{v} \in \mathbf{V}_D$ and $\alpha \in D$. Then*

$$U_\alpha^{\min}(\mathbf{v}) = \text{span}\left\{(\text{id}_{V_\alpha} \otimes \varphi^{[\alpha]})(\mathbf{v}) : \varphi^{[\alpha]} \in (\mathbf{V}_\alpha^{[\alpha]})^*\right\}, \quad (4.2)$$

where $(\text{id}_{V_\alpha} \otimes \varphi^{[\alpha]})(\mathbf{v}) \in V_\alpha$ is the contraction of $\mathbf{v}$ with the functional $\varphi^{[\alpha]}$ in all directions except α .

Proof. Write $\mathbf{v} = \sum_\nu u_\nu \otimes \mathbf{U}_\nu$ with $u_\nu \in V_\alpha$ and $\mathbf{U}_\nu \in \mathbf{V}_\alpha^{[\alpha]}$. For $\varphi \in (\mathbf{V}_\alpha^{[\alpha]})^*$: $(\text{id}_{V_\alpha} \otimes \varphi)(\mathbf{v}) = \sum_\nu \varphi(\mathbf{U}_\nu) u_\nu \in \text{span}\{u_\nu\} = U_\alpha^{\min}(\mathbf{v})$. Conversely, by taking $\varphi = \varphi_\nu$ dual to a basis of $U_{D \setminus \{\alpha\}}^{\min}(\mathbf{v})$, one recovers a basis of $U_\alpha^{\min}(\mathbf{v})$, so equality holds. $\square$

Remark 4.1.6. In Theorem 4.1.5, one can replace $(\mathbf{V}_\alpha^{[\alpha]})^*$ by $(U_{D \setminus \{\alpha\}}^{\min}(\mathbf{v}))^*$ without changing the span, since $\mathbf{v} \in V_\alpha \otimes_a U_{D \setminus \{\alpha\}}^{\min}(\mathbf{v})$. This gives the tighter formula

$$U_\alpha^{\min}(\mathbf{v}) = \text{span}\left\{(\text{id}_{V_\alpha} \otimes \varphi^{[\alpha]})(\mathbf{v}) : \varphi^{[\alpha]} \in (U_{D \setminus \{\alpha\}}^{\min}(\mathbf{v}))^*\right\}, \quad (4.3)$$

which is often more useful in practice.

4.2 Lattice properties and inclusion under partitions

Monotonicity

Proposition 4.2.1 (Monotonicity of minimal subspaces). *Let $\mathbf{v} \in \mathbf{V}_D$ and $U_\alpha \in \mathbb{G}_{r_\alpha}(V_\alpha)$ for each $\alpha \in D$. Then:*

$$\mathbf{v} \in {}_a \bigotimes_{\alpha \in D} U_\alpha \implies U_\alpha^{\min}(\mathbf{v}) \subset U_\alpha \quad \forall \alpha \in D.$$

Proof. Under $\mathbf{v} \in {}_a \bigotimes_{\alpha \in D} U_\alpha$, the identification $\mathbf{V}_D \cong V_\alpha \otimes_a \mathbf{V}_\alpha^{[\alpha]}$ gives $\mathbf{v} \in U_\alpha \otimes_a \mathbf{V}_\alpha^{[\alpha]}$, so the minimality of $U_\alpha^{\min}(\mathbf{v})$ (Theorem 4.1.3(b)) yields $U_\alpha^{\min}(\mathbf{v}) \subset U_\alpha$. $\square$

Inclusion under partition refinement

The following proposition is the key algebraic fact underlying the tree-based format: it shows that minimal subspaces at a coarser level of a tree are contained in tensor products of minimal subspaces at finer levels.

Proposition 4.2.2 (Inclusion under refinement). *Let $\mathbf{v} \in \mathbf{V}_D$ and let $\alpha \subset D$ with $\#(\alpha) \geq 2$. Let $\mathcal{P}_\alpha$ be a non-trivial partition of α . Then*

$$U_\alpha^{\min}(\mathbf{v}) \subset {}_a \bigotimes_{\beta \in \mathcal{P}_\alpha} U_\beta^{\min}(\mathbf{v}). \quad (4.4)$$

Proof. From the canonical identification $\mathbf{V}_D \cong \mathbf{V}_\alpha \otimes_a \mathbf{V}_\alpha^{[\alpha]}$ and Theorem 4.1.5,

$$U_\alpha^{\min}(\mathbf{v}) = \text{span} \left\{ (\text{id}_{\mathbf{V}_\alpha} \otimes \varphi^{[\alpha]})(\mathbf{v}) : \varphi^{[\alpha]} \in (U_{\alpha^c}^{\min}(\mathbf{v}))^* \right\}.$$

For each such element $w_\alpha = (\text{id}_{\mathbf{V}_\alpha} \otimes \varphi^{[\alpha]})(\mathbf{v}) \in \mathbf{V}_\alpha$, we apply Proposition 4.2.1 to the tensor space $\mathbf{V}_\alpha \cong {}_a \bigotimes_{\beta \in \mathcal{P}_\alpha} \mathbf{V}_\beta$. Since $w_\alpha \in U_\alpha^{\min}(\mathbf{v})$ and $\mathbf{V}_\alpha = {}_a \bigotimes_{\beta \in \mathcal{P}_\alpha} \mathbf{V}_\beta$, one has $w_\alpha \in {}_a \bigotimes_{\beta \in \mathcal{P}_\alpha} U_\beta^{\min}(w_\alpha)$. Moreover, from the minimality of $U_\beta^{\min}(\mathbf{v})$ and the relation between contractions at different levels, $U_\beta^{\min}(w_\alpha) \subset U_\beta^{\min}(\mathbf{v})$ for each $\beta \in \mathcal{P}_\alpha$. Hence $w_\alpha \in {}_a \bigotimes_{\beta \in \mathcal{P}_\alpha} U_\beta^{\min}(\mathbf{v})$, and taking the span over all such w_α gives (4.4). $\square$

Remark 4.2.3. Proposition 4.2.2 applied iteratively at every internal vertex $\alpha \in T_D$ of a dimension partition tree (Chapter 5) gives the fundamental compatibility of minimal subspaces with the tree structure:

$$U_\alpha^{\min}(\mathbf{v}) \subset {}_a \bigotimes_{\beta \in S(\alpha)} U_\beta^{\min}(\mathbf{v}) \quad \forall \alpha \in T_D \setminus \mathcal{L}(T_D). \quad (4.5)$$

This is the algebraic foundation of the tree-based format in Chapter 6.

New characterisation of minimal subspaces

The following result, proved in [FHN21] and used in the construction of tree-based tensor formats, gives a characterisation of $U_\beta^{\min}(\mathbf{v})$ via contractions of $U_\alpha^{\min}(\mathbf{v})$ for $\alpha \supset \beta$.

Proposition 4.2.4. *Let $\mathbf{v} \in \mathbf{V}_D$, $\alpha \subset D$ with $\#(\alpha) \geq 2$, and $\mathcal{P}_\alpha$ a non-trivial partition of α . Suppose that $\mathbf{V}_\alpha$ and $\mathbf{V}_\beta$ for $\beta \in \mathcal{P}_\alpha$ are normed spaces. Then for each $\beta \in \mathcal{P}_\alpha$:*

$$U_\beta^{\min}(\mathbf{v}) = \text{span} \left\{ (\text{id}_{\mathbf{V}_\beta} \otimes \varphi^{(\alpha \setminus \beta)})(\mathbf{v}_\alpha) : \mathbf{v}_\alpha \in U_\alpha^{\min}(\mathbf{v}), \varphi^{(\alpha \setminus \beta)} \in \left({}_a \bigotimes_{\gamma \in \mathcal{P}_\alpha \setminus \{\beta\}} \mathbf{V}_\alpha[\gamma] \right)^* \right\}. \quad (4.6)$$

Proof. Applying Theorem 4.1.5 twice — first for the full index set D and then for α — one obtains two nested contraction characterisations. The formula (4.6) follows by composing these and using that $\mathbf{v} \in \mathbf{V}_\alpha \otimes_a U_{\alpha^c}^{\min}(\mathbf{v})$ allows the replacement of the ambient dual by that of $U_{\alpha^c}^{\min}(\mathbf{v})$. $\square$

4.3 Admissible ranks and the disjoint decomposition

Admissible Tucker ranks

Recall from Chapter 3 (Definition 3.3.3) that the set of admissible Tucker ranks is

$$\mathcal{AD}(\mathbf{V}_D) = \left\{ (r_\alpha(\mathbf{v}))_{\alpha \in D} \in \mathbb{N}_0^d : \mathbf{v} \in \mathbf{V}_D \right\}.$$

The following proposition characterises which tuples are admissible.

Proposition 4.3.1. *A tuple $\mathbf{r} = (r_\alpha)_{\alpha \in D} \in \mathbb{N}_0^d$ belongs to $\mathcal{AD}(\mathbf{V}_D)$ if and only if there exist subspaces $U_\alpha \in \mathbb{G}_{r_\alpha}(V_\alpha)$ and a tensor $\mathbf{v} \in \mathfrak{M}_{\mathbf{r}}(\otimes_{\alpha \in D} U_\alpha)$, i.e., a tensor whose α -unfolding has rank exactly r_α for all $\alpha \in D$.*

Proposition 4.3.2. *For $\mathbf{r} \in \mathcal{AD}(\mathbf{V}_D)$:*

- (i) $r_\alpha \leq \dim V_\alpha$ for all $\alpha \in D$.
- (ii) $r_\alpha = r_{\alpha^c} := \dim U_{D \setminus \{\alpha\}}^{\min}(\mathbf{v})$ for all $\alpha \in D$ (symmetry of marginal ranks).
- (iii) $r_\alpha \leq \prod_{\beta \neq \alpha} r_\beta$ for all $\alpha \in D$.

Proof. Statements (i) and (iii) follow from the representation formula (3.9) and rank properties of multi-index arrays. Statement (ii) follows from Proposition 4.1.2 applied to $V_\alpha \otimes_a \mathbf{V}_\alpha^{[\alpha]}$. $\square$

The disjoint decomposition

Theorem 4.1.3 immediately gives the following structural result.

Theorem 4.3.3. *The algebraic tensor space $\mathbf{V}_D$ admits the disjoint decomposition*

$$\mathbf{V}_D = \bigsqcup_{\mathbf{r} \in \mathcal{AD}(\mathbf{V}_D)} \mathfrak{M}_{\mathbf{r}}(\mathbf{V}_D). \quad (4.7)$$

Each $\mathfrak{M}_{\mathbf{r}}(\mathbf{V}_D)$ is non-empty for $\mathbf{r} \in \mathcal{AD}(\mathbf{V}_D)$. The zero tensor $\mathbf{0}$ belongs to $\mathfrak{M}_{\mathbf{0}}(\mathbf{V}_D)$ and each elementary tensor $\otimes_\alpha v_\alpha$ (with all $v_\alpha \neq 0$) belongs to $\mathfrak{M}_{\mathbf{1}}(\mathbf{V}_D)$.

Remark 4.3.4 (Topology of the decomposition). While $\mathfrak{M}_{\mathbf{r}}(\mathbf{V}_D)$ is algebraically well-defined, its topological structure is non-trivial. The key result — that each $\mathfrak{M}_{\mathbf{r}}(\mathbf{V}_D)$ is a $\mathcal{C}^\infty$ -Banach manifold immersed in the ambient Banach tensor space $\mathbf{V}_{D, \|\cdot\|_D}$ — is established in Chapter 8. The immersion requires the condition (Inj) on the norm $\|\cdot\|_D$ (Definition 4.4.1).

Stratification by rank

Definition 4.3.5. For $\mathbf{r} \in \mathcal{AD}(\mathbf{V}_D)$, the set of tensors with Tucker rank *bounded by* $\mathbf{r}$ is

$$\mathfrak{M}_{\leq \mathbf{r}}(\mathbf{V}_D) := \bigsqcup_{\substack{\mathbf{s} \in \mathcal{AD}(\mathbf{V}_D) \\ \mathbf{s} \leq \mathbf{r}}} \mathfrak{M}_{\mathbf{s}}(\mathbf{V}_D) = \left\{ \mathbf{v} \in \mathbf{V}_D : r_{\alpha}(\mathbf{v}) \leq r_{\alpha} \ \forall \alpha \in D \right\}, \quad (4.8)$$

where $\mathbf{s} \leq \mathbf{r}$ means $s_{\alpha} \leq r_{\alpha}$ for all $\alpha \in D$.

The set $\mathfrak{M}_{\leq \mathbf{r}}(\mathbf{V}_D)$ is not a linear subspace (it is not closed under addition) but it is closed under scalar multiplication. The key property for applications — that best approximations from $\mathfrak{M}_{\leq \mathbf{r}}(\mathbf{V}_D)$ exist in Banach tensor spaces — is proved in Section 4.4.

4.4 Best approximations in Banach tensor spaces

In numerical applications, one needs to approximate a given tensor $\mathbf{v} \in \mathbf{V}_{D, \|\cdot\|_D}$ by a tensor of bounded rank. The existence of such a best approximation is non-trivial: the set $\mathfrak{M}_{\leq \mathbf{r}}(\mathbf{V}_D)$ is not convex, so general theorems about proximinal sets do not apply. The proof uses the weak closedness of $\mathfrak{M}_{\leq \mathbf{r}}(\mathbf{V}_D)$.

Crossnorms and the injective norm condition

Definition 4.4.1. Let $(V_{\alpha}, \|\cdot\|_{\alpha})$ be normed spaces and $\|\cdot\|_D$ a norm on $\mathbf{V}_D$. We say $\|\cdot\|_D$ is a *crossnorm* if

$$\left\| \bigotimes_{\alpha \in D} v_{\alpha} \right\|_D = \prod_{\alpha \in D} \|v_{\alpha}\|_{\alpha} \quad \forall v_{\alpha} \in V_{\alpha}. \quad (4.9)$$

The *injective norm* $\|\cdot\|_{\varepsilon}$ on $\mathbf{V}_D$ is defined by

$$\|\mathbf{v}\|_{\varepsilon} := \sup \left\{ \left| \left(\bigotimes_{\alpha \in D} \varphi_{\alpha} \right) (\mathbf{v}) \right| : \varphi_{\alpha} \in V_{\alpha}^*, \ \|\varphi_{\alpha}\|_{V_{\alpha}^*} \leq 1 \right\}.$$

We say $\|\cdot\|_D$ satisfies condition (Inj) if $\|\mathbf{v}\|_D \geq \|\mathbf{v}\|_{\varepsilon}$ for all $\mathbf{v} \in \mathbf{V}_D$.

A crossnorm satisfies (Inj) if and only if it is a *reasonable crossnorm* in the sense that its dual norm is also a crossnorm. The injective norm is the smallest reasonable crossnorm (see [Hac19], Chapter 4).

Remark 4.4.2. The L^p norm on $L^p(\prod_{\alpha} I_{\alpha})$ is a reasonable crossnorm for $1 \leq p \leq \infty$ (see [Hac19], Example 4.72). The Sobolev norm $\|\cdot\|_{(0,1),p}$ on $H^{1,p}(I_1) \otimes_a H^{1,p}(I_2)$ is a crossnorm (Example 4.42 in [Hac19]), hence satisfies (Inj).

Weak closedness of bounded-rank sets

Theorem 4.4.3 (Weak closedness, [FH12, Theorem 5.1]). *Let $(V_\alpha, \|\cdot\|_\alpha)$ be normed spaces and let $\|\cdot\|_D$ be a norm on $\mathbf{V}_D$ satisfying condition (Inj). Then the set $\mathfrak{M}_{\leq \mathfrak{r}}(\mathbf{V}_D)$ is weakly closed in $\mathbf{V}_{D, \|\cdot\|_D}$.*

Sketch. Let $(\mathbf{v}_n) \subset \mathfrak{M}_{\leq \mathfrak{r}}(\mathbf{V}_D)$ converge weakly to $\mathbf{v} \in \mathbf{V}_{D, \|\cdot\|_D}$. We must show $\mathbf{v} \in \mathfrak{M}_{\leq \mathfrak{r}}(\mathbf{V}_D)$, i.e., $r_\alpha(\mathbf{v}) \leq r_\alpha$ for all $\alpha \in D$.

Fix $\alpha \in D$ and consider the α -unfolding map $\mathcal{M}_\alpha : \mathbf{V}_{D, \|\cdot\|_D} \rightarrow \mathcal{L}((V_\alpha)^*, \mathbf{V}_\alpha^{[\alpha]})$ defined by $\mathcal{M}_\alpha(\mathbf{v})(\varphi) = (\varphi \otimes \text{id})(\mathbf{v})$. Since $r_\alpha(\mathbf{v}_n) \leq r_\alpha$, we have $\text{rank} \mathcal{M}_\alpha(\mathbf{v}_n) \leq r_\alpha$ for all n . The condition (Inj) implies that $\mathcal{M}_\alpha$ is continuous for the weak topology on $\mathbf{V}_{D, \|\cdot\|_D}$ and the weak operator topology on $\mathcal{L}((V_\alpha)^*, \mathbf{V}_\alpha^{[\alpha]})$. Since the set of operators of rank $\leq r_\alpha$ is closed in the weak operator topology (it is a closed variety), the limit $\mathcal{M}_\alpha(\mathbf{v})$ has rank $\leq r_\alpha$, i.e., $r_\alpha(\mathbf{v}) \leq r_\alpha$. $\square$

Existence of best approximations

Theorem 4.4.4 (Best approximation, [FH12, Theorem 5.3]). *Let $(V_\alpha, \|\cdot\|_\alpha)$ be normed spaces, let $\|\cdot\|_D$ be a norm on $\mathbf{V}_D$ satisfying (Inj), and assume that $\mathbf{V}_{D, \|\cdot\|_D}$ is reflexive. Then for every $\mathbf{u} \in \mathbf{V}_{D, \|\cdot\|_D}$ and $\mathfrak{r} \in \mathcal{AD}(\mathbf{V}_D)$, there exists $\mathbf{v}_* \in \mathfrak{M}_{\leq \mathfrak{r}}(\mathbf{V}_{D, \|\cdot\|_D})$ such that*

$$\|\mathbf{u} - \mathbf{v}_*\|_D = \inf_{\mathbf{v} \in \mathfrak{M}_{\leq \mathfrak{r}}(\mathbf{V}_{D, \|\cdot\|_D})} \|\mathbf{u} - \mathbf{v}\|_D. \quad (4.10)$$

Proof. Since $\mathbf{V}_{D, \|\cdot\|_D}$ is reflexive, any bounded sequence has a weakly convergent subsequence (Eberlein–Šmulian theorem). Let $(\mathbf{v}_n) \subset \mathfrak{M}_{\leq \mathfrak{r}}(\mathbf{V}_{D, \|\cdot\|_D})$ be a minimising sequence for the infimum in (4.10). The sequence $(\mathbf{v}_n)$ is bounded (the infimum is attained or approached by a bounded set), so by reflexivity there is a weakly convergent subsequence $\mathbf{v}_{n_k} \rightharpoonup \mathbf{v}_*$. By Theorem 4.4.3, $\mathbf{v}_* \in \mathfrak{M}_{\leq \mathfrak{r}}(\mathbf{V}_{D, \|\cdot\|_D})$. Since the norm is weakly lower semicontinuous, $\|\mathbf{u} - \mathbf{v}_*\|_D \leq \liminf_k \|\mathbf{u} - \mathbf{v}_{n_k}\|_D = \inf$, so $\mathbf{v}_*$ is a best approximation. $\square$

Corollary 4.4.5. *For $I_\alpha \subset \mathbb{R}$ ($\alpha \in D$) and $1 < p < \infty$, the Banach tensor space $L^p(\prod_\alpha I_\alpha)$ is reflexive and its norm satisfies (Inj) (Example 4.4.2). Hence every $f \in L^p(\prod_\alpha I_\alpha)$ has a best Tucker approximation of any prescribed rank.*

Corollary 4.4.6. *For $1 < p < \infty$, $N \geq 1$, and bounded intervals I_α :*

- (i) $H^{N,p}(\prod_\alpha I_\alpha)$ with the standard Sobolev norm is reflexive and satisfies (Inj).
- (ii) The space $H^{1,p}(I_1) \otimes_{\|\cdot\|_{(0,1),p}} H^{1,p}(I_2)$ with the partial-derivative crossnorm satisfies (Inj).

In both cases, best Tucker approximations of any rank exist.

Example 4.4.7. Consider $d = 2$, $V_1 = H^1(0, 1)$, $V_2 = H^1(0, 1)$, and the ambient space $V_1 \otimes_{H^1} V_2$ with the norm $\|f\|_{H^1}^2 = \|f\|_{L^2}^2 + \|\partial_{x_2} f\|_{L^2}^2$. This Banach tensor space is reflexive and satisfies (Inj), so every $f \in V_1 \otimes_{H^1} V_2$ has a best approximation of rank $\leq r$ for any $r \geq 1$. This setting arises in parametric PDEs where x_1 is the spatial variable and x_2 is the stochastic parameter.

Extension to intersections

In many applications (including the tree-based format), the ambient space is an intersection of Banach tensor spaces, each with its own norm.

Theorem 4.4.8 (Best approximation in intersections, [FH12, Theorem 5.6]). *Let $\mathbf{V}_{D, \|\cdot\|_D}^{(1)}, \dots, \mathbf{V}_{D, \|\cdot\|_D}^{(k)}$ be reflexive Banach tensor spaces over the same algebraic tensor space $\mathbf{V}_D$, each with a norm satisfying (Inj). Equip the intersection $Z = \mathbf{V}_{D, \|\cdot\|_D}^{(1)} \cap \dots \cap \mathbf{V}_{D, \|\cdot\|_D}^{(k)}$ with the norm $\|\mathbf{v}\|_Z = \max_i \|\mathbf{v}\|_i$. If Z is reflexive (which holds, e.g., when $k = 2$ and both norms are reflexive), then every $\mathbf{u} \in Z$ has a best approximation in $\mathfrak{M}_{\leq r}(Z)$.*

4.5 Historical notes and comparison with the literature

The notion of minimal subspace was introduced in [FH12]. It generalises to infinite dimensions the classical concept of matrix rank: for $d = 2$ and finite-dimensional V_α , the α -rank $r_\alpha(\mathbf{v}) = \text{rank} \mathcal{M}_\alpha(\mathbf{v})$ coincides with the classical matrix rank of the α -unfolding, introduced by Hitchcock [Hit27] in 1927. The higher-order version for finite-dimensional spaces was studied by Tucker [Tuc66], De Lathauwer et al. [DLDMV00], and Kolda–Bader [KB09].

The infinite-dimensional theory of [FH12] is motivated by the numerical treatment of high-dimensional PDEs, where $V_\alpha = L^2(\mathbb{R}^3)$ or Sobolev spaces, and the ambient tensor space is infinite-dimensional. The existence of best approximations (Theorem 4.4.4) removes a fundamental obstacle in the convergence analysis of alternating least-squares and related iterative methods.

The weak closedness argument of Theorem 4.4.3 is modelled on Uschmajew’s result for the case of the spectral norm on finite-dimensional spaces [UV13]. The innovation of [FH12] is the extension to general reflexive Banach spaces under the crossnorm condition (Inj), which is the natural condition for tensor product norms.

The connection between minimal subspaces and the geometry of Tucker manifolds is developed in Chapter 8, where the rank map $\varrho : \mathbf{V}_D \rightarrow \prod_\alpha \mathbb{G}_{r_\alpha}(V_\alpha)$ defined by $\varrho(\mathbf{v}) = (U_\alpha^{\min}(\mathbf{v}))_{\alpha \in D}$ plays a central role in the construction of local charts.

Part II

Tree-Based Tensor Formats

Chapter 5

Dimension Partition Trees

The Tucker format developed in Chapters 3 and 4 corresponds to the shallowest possible hierarchical structure: all d factor spaces are placed at the same level below the root. Richer formats arise by organising the dimensions into a *hierarchy*, encoded by a rooted tree. This chapter introduces dimension partition trees, establishes their basic properties, and presents the principal examples (Tucker, Hierarchical Tucker, Tensor Train). The material prepares the ground for the algebraic (Chapter 6) and topological (Chapter 7) development of tree-based tensor formats.

Throughout, $D = \{1, \dots, d\}$ with $d \geq 2$.

5.1 Definition and basic properties

Definition 5.1.1 (Dimension partition tree). A finite rooted tree T_D is called a *dimension partition tree of D* if:

- (a) Every vertex $\alpha \in T_D$ is a non-empty subset of D .
- (b) The root of T_D is D itself.
- (c) Every non-leaf vertex $\alpha \in T_D$ (i.e., every α with $\#(\alpha) \geq 2$) has at least two children, and the set of children of α , denoted $S(\alpha)$, forms a *non-trivial partition* of α (i.e., a partition into at least two non-empty subsets whose union is α).
- (d) Every vertex α with $\#(\alpha) = 1$ has no children; such vertices are called *leaves*.

The set of all leaves of T_D is denoted $\mathcal{L}(T_D)$. Condition (c) forces every non-leaf to have at least two children, so the tree is never a path of singletons. The following observation is immediate from the definition.

Proposition 5.1.2. *For any dimension partition tree T_D , the set of leaves coincides with the set of singletons of D :*

$$\mathcal{L}(T_D) = \{\{j\} : j \in D\}.$$

In particular, $\mathcal{L}(T_D)$ is the trivial partition of D into singletons.

Proof. By (d), leaves are exactly the vertices with $\#(\alpha) = 1$, i.e., the singletons $\{j\}$ for $j \in D$. Since the root is D and each non-leaf splits into a partition of its label set, every $j \in D$ is eventually reached as a leaf by iterating the partition. $\square$

Levels, depth, and level-partitions

Definition 5.1.3. The *level* of a vertex $\alpha \in T_D$ is defined recursively by

$$\text{level}(D) = 0, \quad \text{level}(\beta) = \text{level}(\alpha) + 1 \quad \text{for } \beta \in S(\alpha).$$

The *depth* of the tree is

$$\text{depth}(T_D) := \max_{\alpha \in T_D} \text{level}(\alpha). \quad (5.1)$$

Since leaves are the only vertices with $\#(\alpha) = 1$, and every path from root to leaf passes through a strictly decreasing chain of subset sizes, the depth satisfies $1 \leq \text{depth}(T_D) \leq d - 1$.

Definition 5.1.4. For $1 \leq k \leq \text{depth}(T_D)$, the *k-th level partition* of D induced by T_D is

$$\mathcal{P}_k(T_D) := \{\alpha \in T_D : \text{level}(\alpha) = k\}. \quad (5.2)$$

By convention, $\mathcal{P}_0(T_D) = \{D\}$.

Proposition 5.1.5. For each $1 \leq k \leq \text{depth}(T_D)$, the collection $\mathcal{P}_k(T_D)$ is a partition of D . Moreover:

(i) Every element of $\mathcal{P}_k(T_D)$ is a subset of some element of $\mathcal{P}_{k-1}(T_D)$ (refinement property).

(ii) $\mathcal{P}_{\text{depth}(T_D)}(T_D) = \mathcal{L}(T_D)$, the trivial partition of D .

Proof. The elements of $\mathcal{P}_k(T_D)$ are precisely the vertex labels at level k . Since each such vertex α is the child of a unique parent $\beta \in \mathcal{P}_{k-1}(T_D)$ satisfying $\alpha \subset \beta$, and since $S(\beta)$ partitions β , the elements of $\mathcal{P}_k(T_D)$ at level k partition D by induction on k . Parts (i) and (ii) follow directly. $\square$

The sequence of partitions $\mathcal{P}_0(T_D) \succ \mathcal{P}_1(T_D) \succ \cdots \succ \mathcal{P}_{\text{depth}(T_D)}(T_D) = \mathcal{L}(T_D)$ (where $\mathcal{P} \succ \mathcal{Q}$ means $\mathcal{Q}$ refines $\mathcal{P}$) captures the hierarchical structure of the tree. This is the central object connecting dimension partition trees to tensor spaces: for each level k , the identification $\mathbf{V}_D \cong \bigotimes_{\alpha \in \mathcal{P}_k(T_D)} \mathbf{V}_\alpha$ (Equation (3.4)) allows one to view $\mathbf{v} \in \mathbf{V}_D$ as a Tucker tensor at level k .

5.2 Special trees and standard formats

The following three families of dimension partition trees correspond to the three most widely used tensor formats.

The Tucker tree

Example 5.2.1 (Tucker tree). The *Tucker tree* T_D^{Tucker} over D has depth 1 and satisfies

$$T_D^{\text{Tucker}} = \{D, \{1\}, \{2\}, \dots, \{d\}\}, \quad S(D) = \mathcal{L}(T_D)[T_D^{\text{Tucker}}] = \{\{j\} : j \in D\}.$$

The only non-trivial partition step is $D \rightarrow \{\{1\}, \dots, \{d\}\}$.

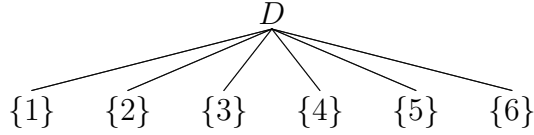


Figure 5.1: Tucker tree over $D = \{1, 2, 3, 4, 5, 6\}$: depth 1, one level partition $\mathcal{P}_1 = \mathcal{L}(T_D)$.

The Tucker tree is the unique tree of depth 1 over D . The corresponding tensor format is exactly the Tucker format of Chapter 3: $\mathcal{FT}_r(\mathbf{V}_D, T_D^{\text{Tucker}}) = \mathfrak{M}_r(\mathbf{V}_D)$.

Binary trees and the Hierarchical Tucker format

Definition 5.2.2. A dimension partition tree T_D is *binary* if every non-leaf vertex has exactly two children: $\#(S(\alpha)) = 2$ for all $\alpha \in T_D$ with $\#(\alpha) \geq 2$.

Binary trees are the natural hierarchical structure for the *Hierarchical Tucker* (HT) format introduced by Hackbusch and Kühn [?].

Example 5.2.3 (Hierarchical Tucker tree). For $D = \{1, 2, 3, 4\}$, the balanced binary HT tree is

$$T_D^{\text{HT}} = \{D, \{1, 2\}, \{3, 4\}, \{1\}, \{2\}, \{3\}, \{4\}\},$$

with $S(D) = \{\{1, 2\}, \{3, 4\}\}$, $S(\{1, 2\}) = \{\{1\}, \{2\}\}$, $S(\{3, 4\}) = \{\{3\}, \{4\}\}$. Here $\text{depth}(T_D^{\text{HT}}) = 2$, with level partitions $\mathcal{P}_1 = \{\{1, 2\}, \{3, 4\}\}$ and $\mathcal{P}_2 = \mathcal{L}(T_D) = \{\{1\}, \{2\}, \{3\}, \{4\}\}$.

Example 5.2.4 (Non-binary tree). For $D = \{1, 2, 3, 4, 5, 6\}$:

$$T_D = \{D, \{1, 2, 3\}, \{4, 5\}, \{6\}, \{1\}, \{2, 3\}, \{4\}, \{5\}, \{2\}, \{3\}\},$$

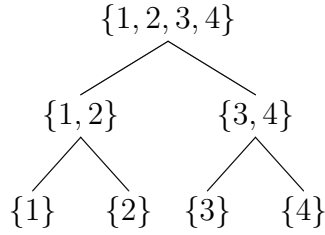


Figure 5.2: Balanced binary tree over $D = \{1, 2, 3, 4\}$: depth 2, corresponding to the HT format.

where $S(D) = \{\{1, 2, 3\}, \{4, 5\}, \{6\}\}$ (three children), $S(\{1, 2, 3\}) = \{\{1\}, \{2, 3\}\}$, $S(\{2, 3\}) = \{\{2\}, \{3\}\}$, $S(\{4, 5\}) = \{\{4\}, \{5\}\}$. Here $\text{depth}(T_D) = 3$, with $\mathcal{P}_1 = \{\{1, 2, 3\}, \{4, 5\}, \{6\}\}$, $\mathcal{P}_2 = \{\{1\}, \{2, 3\}, \{4\}, \{5\}, \{6\}\}$, $\mathcal{P}_3 = \mathcal{L}(T_D)$.

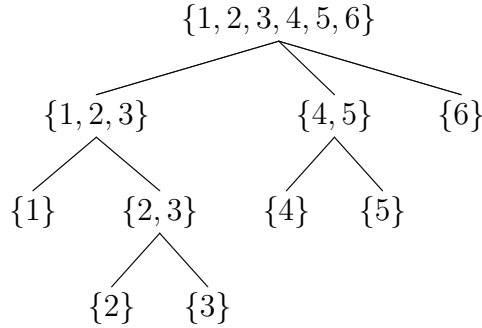


Figure 5.3: Non-binary dimension partition tree over $D = \{1, 2, 3, 4, 5, 6\}$: depth 3.

The Tensor Train tree

Example 5.2.5 (Tensor Train tree). For $D = \{1, \dots, d\}$, $d \geq 2$, the *Tensor Train* (TT) tree [?] is the linear (chain) binary tree

$$T_D^{\text{TT}} = \{D\} \cup \{\{j\} : 1 \leq j \leq d\} \cup \{\{1, \dots, j\} : 2 \leq j \leq d-1\},$$

with $S(\{1, \dots, j\}) = \{\{1, \dots, j-1\}, \{j\}\}$ for $j \geq 2$. Here $\text{depth}(T_D^{\text{TT}}) = d-1$. The level partitions are:

$$\mathcal{P}_k(T_D^{\text{TT}}) = \{\{1, \dots, d-k\}, \{d-k+1\}, \dots, \{d\}\}, \quad 1 \leq k \leq d-1.$$

Remark 5.2.6. The TT format is a special case of the HT format: T_D^{TT} is a binary tree with maximal depth $d-1$ and a linear (path) structure, whereas HT trees are typically balanced with depth $\lceil \log_2 d \rceil$. This is reflected in their computational cost: TT has storage $O(dr^2n)$ while balanced HT has $O(dr^3)$, with r the maximal rank and $n = \max_j \dim V_j$ [?, ?].

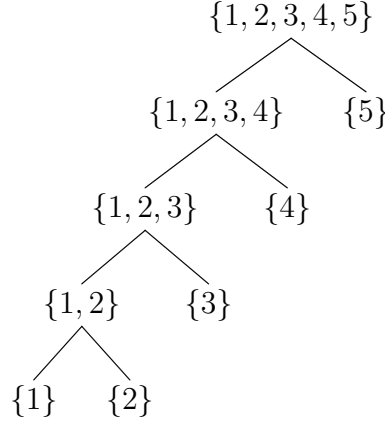


Figure 5.4: Tensor Train tree over $D = \{1, 2, 3, 4, 5\}$: depth 4, linear binary structure.

5.3 Graph-theoretic properties

We record here the combinatorial facts about dimension partition trees that will be used in Chapters 6 and 9.

Proposition 5.3.1. *Let T_D be a dimension partition tree over $D = \{1, \dots, d\}$. Then:*

- (i) $\#(\mathcal{L}(T_D)) = d$. The number of internal (non-leaf) vertices satisfies $\#(T_D \setminus \mathcal{L}(T_D)) \leq d - 1$.
- (ii) If T_D is binary, then $\#(T_D) = 2d - 1$, with exactly $d - 1$ internal vertices.
- (iii) $\text{depth}(T_D) \leq d - 1$, with equality if and only if $T_D = T_D^{\text{TT}}$ (up to re-labelling).

Proof. Part (i): every leaf is a singleton $\{j\}$ for a unique $j \in D$, so $\#(\mathcal{L}(T_D)) = d$. The bound on internal vertices follows since each internal vertex contributes at least two children which are eventually singletons. Part (ii): in a binary tree with d leaves, the number of internal nodes is exactly $d - 1$, giving $2d - 1$ total vertices. Part (iii): the depth is maximised when each step reduces the largest set by exactly one element, giving the TT structure. $\square$

Proposition 5.3.2 (Recursive subtree structure). *Let T_D be a dimension partition tree over D and $\alpha \in T_D$ with $\#(\alpha) \geq 2$. Then the restriction*

$$T_D|_\alpha := \{\beta \in T_D : \beta \subseteq \alpha\}$$

is itself a dimension partition tree over α (with root α).

This recursive structure is fundamental to the inductive proofs in Chapter 6: properties of tree-based tensor formats are proved by induction on $\text{depth}(T_D)$, reducing at each step to the subtrees $T_D|_\beta$ for $\beta \in S(D)$.

5.4 The intersection representation

The central structural theorem of the tree-based theory expresses the set of tree-based tensors with fixed rank as an intersection of Tucker manifolds — one at each level of the tree. This is the key result connecting the tree structure to the geometry of Chapters 9.

For each level partition $\mathcal{P}_k(T_D)$, recall from Section 3.3 that the identification $\mathbf{V}_D \cong \bigotimes_{\alpha \in \mathcal{P}_k(T_D)} \mathbf{V}_\alpha$ allows one to define the Tucker set

$$\mathfrak{M}_{\mathbf{r}_k}(\mathbf{V}_D, \mathcal{P}_k(T_D)) := \left\{ \mathbf{v} \in \mathbf{V}_D : \dim U_\alpha^{\min}(\mathbf{v}) = r_\alpha \text{ for all } \alpha \in \mathcal{P}_k(T_D) \right\}, \quad (5.3)$$

where $\mathbf{r}_k = (r_\alpha)_{\alpha \in \mathcal{P}_k(T_D)}$.

Theorem 5.4.1 (Intersection representation, [?, Theorem 3.12]). *Let T_D be a dimension partition tree over D and $\mathbf{r} = (r_\alpha)_{\alpha \in T_D} \in \mathcal{AD}(\mathbf{V}_D, T_D)$. Then*

$$\mathcal{FT}_{\mathbf{r}}(\mathbf{V}_D, T_D) = \bigcap_{k=1}^{\text{depth}(T_D)} \mathfrak{M}_{\mathbf{r}_k}(\mathbf{V}_D, \mathcal{P}_k(T_D)), \quad (5.4)$$

where $\mathbf{r}_k = (r_\alpha)_{\alpha \in \mathcal{P}_k(T_D)}$.

Sketch. The inclusion $\mathcal{FT}_{\mathbf{r}}(\mathbf{V}_D, T_D) \subseteq \bigcap_k \mathfrak{M}_{\mathbf{r}_k}(\mathbf{V}_D, \mathcal{P}_k(T_D))$ is immediate: if $\dim U_\alpha^{\min}(\mathbf{v}) = r_\alpha$ for all $\alpha \in T_D$, then in particular for all $\alpha \in \mathcal{P}_k(T_D)$ at each level k . For the reverse inclusion, one uses Proposition 4.2.2 inductively: from $\dim U_\alpha^{\min}(\mathbf{v}) = r_\alpha$ at the leaf level ($k = \text{depth}(T_D)$) and the compatibility condition $U_\alpha^{\min}(\mathbf{v}) \subset {}_a \bigotimes_{\beta \in S(\alpha)} U_\beta^{\min}(\mathbf{v})$, one deduces $\dim U_\alpha^{\min}(\mathbf{v}) = r_\alpha$ at all intermediate levels. See [?] for the complete proof. $\square$

Remark 5.4.2. For $T_D = T_D^{\text{Tucker}}$ (depth 1), the intersection (??) reduces to a single factor: $\mathcal{FT}_{\mathbf{r}}(\mathbf{V}_D, T_D^{\text{Tucker}}) = \mathfrak{M}_{\mathbf{r}_1}(\mathbf{V}_D, \mathcal{L}(T_D)) = \mathfrak{M}_{\mathbf{r}}(\mathbf{V}_D)$, recovering the Tucker format.

Example 5.4.3. For $D = \{1, 2, 3, 4\}$ and the balanced HT tree T_D^{HT} of Example ?? with $\mathbf{r} = (r_D, r_{\{1,2\}}, r_{\{3,4\}}, r_{\{1\}}, r_{\{2\}}, r_{\{3\}}, r_{\{4\}})$:

$$\mathcal{FT}_{\mathbf{r}}(\mathbf{V}_D, T_D^{\text{HT}}) = \mathfrak{M}_{\mathbf{r}_1}(\mathbf{V}_D, \mathcal{P}_1) \cap \mathfrak{M}_{\mathbf{r}_2}(\mathbf{V}_D, \mathcal{P}_2),$$

where $\mathcal{P}_1 = \{\{1, 2\}, \{3, 4\}\}$ and $\mathcal{P}_2 = \mathcal{L}(T_D)$. Concretely:

$$\begin{aligned} \mathfrak{M}_{\mathbf{r}_1}(\mathbf{V}_D, \mathcal{P}_1) &= \left\{ \mathbf{v} \in \mathbf{V}_D : \dim U_{\{1,2\}}^{\min}(\mathbf{v}) = r_{\{1,2\}}, \dim U_{\{3,4\}}^{\min}(\mathbf{v}) = r_{\{3,4\}} \right\}, \\ \mathfrak{M}_{\mathbf{r}_2}(\mathbf{V}_D, \mathcal{P}_2) &= \left\{ \mathbf{v} \in \mathbf{V}_D : \dim U_{\{j\}}^{\min}(\mathbf{v}) = r_{\{j\}}, j = 1, 2, 3, 4 \right\}. \end{aligned}$$

The intersection captures tensors that are simultaneously Tucker at the level of pairs ($\{1, 2\}, \{3, 4\}$) and at the level of singletons.

Remark 5.4.4 (Degenerate case). Not all intersections define distinct sets. If for some internal vertex $\alpha \in T_D$ the rank r_α equals $\prod_{\beta \in S(\alpha)} r_\beta$, the rank constraint at α is

automatically satisfied whenever the constraints at the children are. In such cases one of the Tucker factors in (??) is redundant. This situation is analysed in detail in [?, Theorem 3.12].

5.5 Historical notes

Dimension partition trees formalise the hierarchical structure of tensor representations that had appeared in various forms in the literature. The HT format was introduced by Hackbusch and Kühn in [?] using a binary tree structure, with the specific aim of reducing the exponential storage of full Tucker tensors to $O(dr^3)$. The TT format, independently introduced by Vidal [?] in quantum information (under the name *matrix product states*) and by Oseledets [?] in numerical mathematics (as *TT decomposition*), corresponds to the linear chain binary tree.

The general notion of a dimension partition tree, unifying these formats under a common combinatorial framework, was introduced in [FHN21]. The intersection representation (Theorem ??) was established in [?] and is the key to the geometric theory of Chapter 9: it reduces the analysis of the tree-based manifold $\mathcal{FT}_t(\mathbf{V}_D, T_D)$ to the intersection of Tucker manifolds, each of which has the structure established in Chapter 8.

The connection between dimension partition trees and quantum physics is deeper than a mere formal analogy. In quantum information, a tensor network is a collection of tensors connected by contraction along shared indices, and the tree structure encodes the entanglement structure of a quantum state. The area law for entanglement entropy in one-dimensional quantum systems [?] is the physical motivation for the TT format; balanced binary trees encode area laws in higher-dimensional systems. This connection is discussed further in Chapter 11.

Chapter 6

Algebraic Tree-Based Tensor Formats

To be written

This chapter is a placeholder. Content will be added in subsequent revisions.

Chapter 7

Topological Tree-Based Tensor Spaces

To be written

This chapter is a placeholder. Content will be added in subsequent revisions.

Part III

Differential Geometry of Tensor Formats

Chapter 8

The Tucker Manifold

To be written

This chapter is a placeholder. Content will be added in subsequent revisions.

Chapter 9

The Tree-Based Manifold

To be written

This chapter is a placeholder. Content will be added in subsequent revisions.

Part IV

Variational Principles and
Applications

Chapter 10

The Dirac–Frenkel Variational Principle

To be written

This chapter is a placeholder. Content will be added in subsequent revisions.

Chapter 11

Open Problems and Perspectives

To be written

This chapter is a placeholder. Content will be added in subsequent revisions.

Appendix A

Banach Manifolds: A Summary

To be written

This chapter is a placeholder. Content will be added in subsequent revisions.

Appendix B

Functional Analysis: Selected Results

To be written

This chapter is a placeholder. Content will be added in subsequent revisions.

Appendix C

Index of Notation

To be written

This chapter is a placeholder. Content will be added in subsequent revisions.

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